

OPEN QUANTUM SAFE

Community Health & Contribution Analysis

liboqs · oqs-provider

Analysis date: March 14, 2026 | Data through: March 2026 | Period covered: 2016–2026

liboqs Health Score 6.7 / 10 Maturing · Broadening	oqs-provider Health Score 4.3 / 10 Concentrated · Declining	Combined Ecosystem 5.8 / 10 Critical Juncture
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Scores assess community breadth, retention, growth trend, and bus factor — not code quality.

Executive Summary

The OQS ecosystem sits at a pivotal moment. liboqs has achieved genuine community momentum — 104 new contributors in 2024 alone and 40 consecutive months of broad engagement — while oqs-provider shows alarming concentration and a sharp post-NIST-standardization decline that demands urgent action.

liboqs — Strengths

- 426 total contributors over 10 years
- 40 consecutive months ≥ 10 active contributors
- 104 new contributors in 2024 (record year)
- Bus factor of 3 on a 4-year window excluding departed contributors (previously reported as 4, which included SWilson4's historical activity) — current active core is 8 people in the Core tier
- Strong recent breadth: 31–37 contributors/month Jun–Aug 2025

oqs-provider — Concerns

- Single contributor (baentsch) holds 45.6% of all activity
- Bus factor of 2 — critical single-point-of-failure
- Activity down 54% and contributor breadth down 51% vs prior year
- 56% of contributors appear only once and never return
- 17 dormant months (< 5 active contributors)

Three urgent priorities: (1) reduce oqs-provider's maintainer concentration before burnout strikes; (2) convert the wave of 2024–2025 new contributors from one-time visitors into recurring participants; (3) bridge contributors between the two repos more systematically.

1. Dataset & Methodology

This report is derived from raw GitHub interaction events exported from the oqs-provider and liboqs repositories. Events include issue comments, pull request review comments, PR review verdicts (approve / request changes / comment), PR lifecycle events (opened, merged, closed), discussion threads and replies, and commit comments. Commits themselves are not counted; the analysis focuses exclusively on community engagement — the conversational and review layer that reflects collaborative health.

Repository	Period	Total Events	Contributors	Months
open-quantum-safe/liboqs	Aug 2016 – Mar 2026	17,472	426	116
open-quantum-safe/oqs-provider	Feb 2021 – Mar 2026	5,310	210	62
Combined ecosystem	Aug 2016 – Mar 2026	22,782	535*	116

** 81 contributors are active in both repositories; 535 is the deduplicated total.*

2. liboqs — Detailed Analysis

2.1 Key Health Metrics

All metrics use a 4-year window (Jan 2022 – Mar 2026), excluding bots and departed contributors (SWilson4, praveksharma, Martyrshot), for a current rather than historical view of project health.

Metric	Value	Context
Total interactions	9,011	4-year window (2022+), excl. departed & bots
Unique contributors	330	4-year window; 80 (24%) appear only once
Bus factor	3	3 people = 50% of activity (excl. departed SWilson4)
Top contributor share	28.2%	baentsch leads 4-year window (dstebila 17.5%)
Top-5 share	60.3%	Moderate concentration — top 5 unchanged in character
One-time contributors	24% (80)	Much lower in 4-year window — many historics were one-time
Retained ≥3 months	29% (123)	Good base of recurring participants
Peak month	Jan 2024	567 events — NIST standardization peak (within 4-year window)
Active streak	40 months	Consecutive months ≥10 contributors
Event trend (yr/yr)	−15.8%	Mild post-peak normalization expected
Breadth trend (yr/yr)	−9.4%	Slight contraction, not yet alarming

2.2 Growth Narrative — Four Eras

Era 1 — Origins (2016–2019): The project emerged with a committed but tiny founding team. Activity was sparse and dominated by 3–5 people. The 2018 peaks (335 events in September) reflect early algorithmic debates as NIST's post-quantum competition gathered momentum. Only 8 of the first 61 contributors (13%) were still active a year later, typical of early open-source projects.

Era 2 — COVID Surge (2020–2021): Remote-work patterns and heightened attention to cryptographic infrastructure drove a step-change. Monthly contributor counts rose from single digits to 10–15, and the first true community structure emerged. New contributor intake in 2021 reached 28, the best retention rate of any cohort (36% active ≥3 months).

Era 3 — NIST Countdown (2022–2024): The approach of NIST's final PQC standards transformed the project. Contributor intake exploded: 38 new in 2022, 71 in 2023, a record 104 in 2024. The January 2024 peak (567 events, 22 contributors) coincided with intense discussion

around the NIST FIPS drafts. The August 2024 high (37 contributors) captures the final push before standardization.

Era 4 — Post-Standardization (2025–present): Following the NIST FIPS publication, activity moderated — expected for a maturing project. Crucially, breadth remained strong: June–September 2025 saw 27–37 contributors per month, higher than any pre-2024 figure. This suggests the community has genuinely grown rather than simply attracted temporary attention.

2.3 Monthly Activity — Last 18 Months

Month	Contributors	Events	Activity bar
2024-10	34	479	
2024-11	26	390	
2024-12	22	146	
2025-01	35	341	
2025-02	31	320	
2025-03	23	205	
2025-04	14	167	
2025-05	25	124	
2025-06	31	512	← peak
2025-07	34	398	
2025-08	37	312	
2025-09	27	347	
2025-10	25	268	
2025-11	22	225	
2025-12	15	87	
2026-01	35	258	
2026-02	28	260	
2026-03*	14	171	← partial

* March 2026 is a partial month at time of data export.

2.4 Key Contributors

Contributor	Events	Months Active	Share	Cross-repo?
dtebila	3,587	114 / 116	20.5%	Yes (minor)
baentsch	3,341	71	19.1%	Yes (major)
SWilson4	1,357	25	7.8%	Departed Jul 2025
christianpaquin	793	75	4.5%	Minor

bhess	746	59	4.3%	Yes
ashman-p	388	35	2.2%	Yes
xvzcf	382	32	2.2%	No
praveksharma	307	26	1.8%	Departed Aug 2025
aidenfoxivey	290	14	1.7%	No ← rising
Martyrshot	287	33	1.6%	Minor

2.5 Cohort Retention Analysis

The cohort table shows the 'stickiness' of each intake year — what fraction of contributors who first appeared in a given year were still active the following year. This is the single most important indicator of community health: you can attract contributors, but you need to retain them.

First year	New contributors	Active yr+1	Active yr+2	≥3 months active
2016	8	7 (88%)	4 (50%)	7 (88%)
2017	21	4 (19%)	1 (5%)	6 (29%)
2018	14	4 (29%)	3 (21%)	7 (50%)
2019	18	5 (28%)	3 (17%)	7 (39%)
2020	27	9 (33%)	3 (11%)	10 (37%)
2021	28	7 (25%)	9 (32%)	10 (36%)
2022	38	6 (16%)	2 (5%)	6 (16%) ⚠
2023	71	23 (32%)	14 (20%)	25 (35%)
2024	104	24 (23%)	8 (8%)	23 (22%) ⚠
2025	84	12 (14%)	—	21 (25%)

The 2022 cohort stands out as particularly weak — only 16% were active for ≥3 months, the worst retention since 2017. This may reflect that contributors attracted during the 'build-up' phase had more transactional (rather than intrinsic) motivations. The 2024 cohort's 23% three-month retention is better than 2022 but still below the 30–36% range achieved in 2019–2021, suggesting onboarding friction remains a problem.

3. oqs-provider — Detailed Analysis

3.1 Key Health Metrics

All metrics use a 4-year window (Jan 2022 – Mar 2026), excluding bots and departed contributors.

Metric	Value	Context
Total interactions	4,704	4-year window (2022+), excl. departed & bots
Unique contributors	204	4-year window; 44 (22%) appear only once
Bus factor	2	Only 2 people = 50% of activity — CRITICAL
Top contributor share	48.5%	baentsch — even higher in 4-year window
Top-3 share	59.5%	baentsch + RodriM11 + dstebila dominate
One-time contributors	56% (117)	Similar to liboqs — same funnel problem
Retained ≥3 months	27% (57)	Comparable retention to liboqs
Peak month	Aug 2024	273 events, 37 contributors
Event trend (yr/yr)	−54.0%	Severe decline — requires intervention
Breadth trend (yr/yr)	−51.3%	Contributor breadth halved year-over-year
Active streak ≥10	17 months	Feb 2023 – Jun 2024 — now broken

3.2 The Concentration Problem

baentsch accounts for 48.5% of oqs-provider engagement over the last 4 years — 2,281 of 4,704 events (excluding departed contributors). This is not a healthy level of core contributor reliance; it is a critical single-point-of-failure. The next two contributors (bhess and dstebila) bring the top-3 share to 56.5%. Effectively, three people carry the entire project.

The danger is not hypothetical. Contributor burnout is the leading cause of open-source project abandonment. A key maintainer reducing availability by even 30% would, based on these numbers, visibly reduce overall project activity. There is currently no second-tier layer of reviewers who could absorb that capacity.

Contributor	Events	Share	Months active	Role
baentsch	2,421	45.6%	58 / 62	Primary maintainer
bhess	307	5.8%	44	Regular contributor
RodriM11	274	5.2%	12	Recent, active
dstebila	254	4.8%	49	Cross-repo (mainly liboqs)

SWilson4	229	4.3%	22	Departed Jul 2025
mouse07410	172	3.2%	15	Regular contributor
ashman-p	111	2.1%	24	Regular contributor
praveksharma	108	2.0%	12	Departed Aug 2025
feventura	82	1.5%	13	Regular contributor
beldmit	78	1.5%	19	Cross-repo

3.3 Trend Analysis — The Post-NIST Cliff

The August 2024 peak (273 events, 37 contributors) represents the high-water mark — coinciding with the final NIST PQC standardization. Since then, oqs-provider has seen a sharp and sustained decline that contrasts with liboqs's relative stability:

Month	Contributors	Events	Activity bar	
2025-04	14	61		
2025-05	14	78		
2025-06	10	30		← breadth low
2025-07	9	35		
2025-08	11	33		
2025-09	17	143		← event spike
2025-10	9	138		
2025-11	13	90		
2025-12	11	75		
2026-01	12	75		
2026-02	9	134		
2026-03*	7	31		

The September–October 2025 event spikes (143 and 138 events) despite low contributor breadth (17 and 9) suggest a pattern of intensive activity by a very small number of people — likely baentsch alone. This reinforces the concentration concern rather than alleviating it.

4. Cross-Repository Ecosystem Analysis

4.1 Contributor Overlap

81 GitHub accounts appear in both repositories at least once, but the meaningful overlap is considerably smaller. Only 25 contributors are active in both repos across 3+ active months each, and just 19 have 10 or more engagement events in each. The larger figure is inflated by single-event cross-touches — for example, one issue comment on the other repo — that do not represent genuine cross-project participation. The 19 truly dual-active contributors are the real connective tissue of the OQS ecosystem.

Overlap threshold	Count	Notes
Any account in both repos (raw count)	81	Inflated by single-event cross-touches
Meaningful: ≥3 events in each repo	52	Casual cross-participation
Genuine: ≥10 events in each repo	19	True dual-repo contributors
Sustained: ≥3 active months in each repo	25	Long-term cross-project engagement
liboqs only (any activity)	345	Large pool, zero provider engagement
oqs-provider only (any activity)	129	Provider specialists — untapped resource

4.2 Review Culture

Both repos show a similar and healthy distribution of review types. The low changes-requested rate (4% in both) is worth monitoring — it can reflect genuine pre-screening culture (PRs discussed before submission) or reviewer passivity. Comment body size data, now available in this updated cache, adds important context.

Repository	Approved	Changes Requested	Comment-only	Total Reviews
liboqs	1,438 (40%)	131 (4%)	1,994 (56%)	3,563
oqs-provider	362 (36%)	45 (4%)	596 (59%)	1,003

4.2a Approval Pattern Analysis — Inline Comment Volume

Year-by-year inline/approval ratio (proxy for review rigour): The ratio of inline PR review comments to approval verdicts has improved markedly since 2020 and remains healthy. A falling ratio would indicate reviewers approving without reading.

Year	Approvals	Inline comments	Ratio	Assessment
2019	50	40	0.80x	Low
2020	147	69	0.47x	Lowest — bandwidth pressure during rapid

				growth
2021	187	202	1.08x	Recovered
2022	129	176	1.36x	Good
2023	284	533	1.88x	Strong
2024	443	1,193	2.69x	Best year — most rigorous
2025	355	547	1.54x	Healthy
2026*	53	165	3.11x	Very high (partial month)

4.2b Approval Body Size Analysis — Comment Text Evidence

Key methodological note: In this dataset, 74% of approvals carry no body text (ranging from 54% in 2026 to 97% in 2017; the improvement trend is real). Individual reviewer rates vary from 36% (baentsch) to 98% (dstebila, ashman-p). This makes empty% alone a weak discriminator. The meaningful signal comes from combining three independent metrics: (1) empty approval body%, (2) median inline comment length, and (3) changes-requested rate. Reviewers who score poorly on all three simultaneously are the genuine concern.

Comment body sizes are measured after stripping quoted text, code blocks, images, and HTML comments — only original new prose is counted.

Reviewer	Appr .	empt y%	appr med	chg_re q%	inline med	ic med	Assessment
baentsch	376	36%	40c	7.0%	125c	262c	Best combined profile — writes summaries
dstebila	774	98%	0c	3.6%	65c	132c	High empty%; inline substance offsets
SWilson4	183	56%	0c	1.5%	88c	197c	Departed Jul 2025
bhess	122	61%	0c	1.6%	95c	200c	Acceptable — healthy inline size
ashman-p	64	98%	0c	0.0%	27c	167c	⚠ Concern — low inline + 0 chg_req ever
praveksharma	61	74%	0c	0.7%	132c	233c	Departed Aug 2025; inline size ok
christianpaquin	43	67%	0c	6.1%	68c	111c	Acceptable — chg_req rate offsets
xvzcf	41	98%	0c	6.2%	118c	138c	Mixed — high empty but substantive inline
Martyrshot	20	40%	5c	3.8%	90c	232c	Acceptable
xuganyu96	17	53%	0c	22.7%	176c	228c	Best chg_req% — thorough
jschanck	10	80%	0c	3.8%	129c	211c	Acceptable — substantive inline
vsoftco	15	87%	0c	0.0%	88c	108c	Low but small volume

Columns: *appr med* = median chars of review_approved body. *inline med* = median chars of PR inline review comments. *ic med* = median chars of issue/discussion comments. All body text stripped of quotes, code blocks, and images before measurement.

Findings with comment size evidence:

- baentsch stands out as the most engaged approver by the combined metrics: 36% empty approvals (best in class), median 40c summary body, 7% changes-requested rate, and 125c median inline comments. This is the profile of a deeply engaged primary maintainer.
- ashman-p is the clearest concern: 98% empty approvals, median inline comment of only 27c (shortest among all significant reviewers), and 0 changes-requested across 64 approvals and 35 months. All three indicators align. The 27c median inline length suggests comments like 'nit' or 'agreed' rather than substantive code feedback.
- dstebila (project co-founder, 114 months) shows 98% empty approvals but a 65c median inline comment — substantive enough to indicate real engagement when reviewing. The high empty-approval count likely reflects a trust-based merge pattern for routine changes from known contributors.
- xvzcf presents a nuanced picture: 98% empty approvals but 118c median inline comments and 6.2% changes-requested — the pattern of a reviewer who does the work in the diff rather than in the verdict summary. Low concern.
- The year-by-year trend in empty approval bodies is improving: from 97% in 2017 to 54% in 2026, with 2024 at 66% as the recent best. This aligns with the broadening contributor base bringing more review culture diversity.

4.2c Top Reviewers by Inline Comment Substance

Ranked by median inline PR review comment length — the clearest indicator of substantive code engagement. Minimum 10 inline comments for inclusion.

Reviewer	Inline comments	Median length	Avg length	Profile
Frauschi	24	273c	376c	Most substantive per-comment — thorough
RodriM11	52	196c	252c	Strong — rising contributor
xuganyu96	59	176c	232c	High median + highest chg_req%
planetf1	27	151c	360c	Substantive, high variance
praveksharma	74	132c	193c	Departed Aug 2025 — was substantive
jschanck	23	129c	215c	Strong reviewer
feventura	26	129c	217c	Strong reviewer
baentsch	825	125c	186c	Highest volume + substantive — core maintainer
ducnguyen-sb	95	110c	173c	Solid

rtjk	27	109c	184c	Solid
cothan	27	110c	163c	Solid
bhess	225	95c	160c	Good — second highest volume
ashman-p	116	27c	—	⚠ Lowest median — suggests minimal comments

4.3 Top Cross-Ecosystem Reviewers

The top reviewers represent the connective tissue of the OQS community. These are the people whose continued engagement is most critical to both code quality and community cohesion.

Reviewer	Review events	Primary repo
baentsch	2,142	oqs-provider (overwhelmingly)
dstebila	1,627	liboqs
SWilson4	830	liboqs
bhess	544	Both
ashman-p	295	liboqs
praveksharma	212	Both
ducnguyen-sb	190	liboqs
christianpaquin	185	liboqs
Martyrshot	152	liboqs
RodriM11	105	oqs-provider (recent)

5. Recommendations

Recommendations are grounded in current activity data (Oct 2025–Mar 2026). The effective active community is 18 people across the Core and Regular tiers (see Section 6.0). Contributors who have formally departed (SWilson4, praveksharma) and those absent for 6+ months (mouse07410, last seen Sep 2025; aidenfoxivey, last 2 events Oct 2025) are excluded.

5.1 Critical — Address Immediately

P1**Formalise oqs-provider co-maintainership with RodriM11**

baentsch accounts for 38% of all oqs-provider events in the last 6 months. RodriM11 is already a Committer in oqs-provider — the formal co-maintainer pathway is already open. The next step is a Maintainer elevation vote, which requires majority Committer approval. RodriM11 has 194 events in 6 months, 28 review events, a 196c median inline length, and 12 months of history. The structural risk is not the absence of a candidate but the absence of a documented succession process and the gap in RodriM11's liboqs formal role (addressed in G6).

P2**Establish a structured first-contribution pathway for oqs-provider**

56% of provider contributors appear exactly once and do not return. At the current activity level (avg 26 contributors/month across both repos, bot-excluded), every retained contributor matters significantly more than in the 2024 peak. Introduce labelled 'good first issue' tickets specifically in oqs-provider, a provider-specific contribution guide (distinct from liboqs), and a personal follow-up comment from a maintainer on every first PR within 7 days. Active contributors who cross the 3-month threshold have historically shown 4x higher long-term retention.

5.2 High Priority — Address Within 3 Months

P3**Actively cultivate the current second-tier review layer**

The current active review layer beyond baentsch and dstebila is thin but real: xuganyu96 (73 review events, 176c inline median), abhi-dev-engg (72 review events, liboqs), RodriM11 (55 review events across both repos). These three are the natural successors to the review load that SWilson4 and praveksharma carried. Direct engagement — assigning them as requested reviewers on specific PR categories, acknowledging their contributions publicly — has strong evidence for accelerating the transition from contributor to maintainer.

P4**Bridge liboqs contributors into oqs-provider**

Of the 18 Core/Regular contributors currently active, 8 participate in both repos. xuganyu96 (152 events in liboqs, 0 in provider) and abhi-dev-engg (109 events in liboqs, 0 in provider) are the highest-priority bridge targets — they clearly have the technical depth. A single targeted Discussion post in oqs-provider with specific labelled issues tailored to liboqs contributors would reach exactly the right audience.

P5

Implement a lightweight monthly health pulse

A one-page automated summary published monthly to GitHub Discussions — active contributor count, new contributors, review throughput, median time-to-first-response — costs one hour to automate and creates lasting accountability. The departure of SWilson4 and praveksharma in mid-2025 would have been visible months earlier with this kind of tracking, allowing proactive response rather than reactive discovery.

5.3 Medium Term — Address Within 6 Months

P6

Formalise mentorship for high-potential current contributors

Four contributors show the profile and recency to become long-term maintainers: RodriM11 (both repos, 12 months, strong review quality), xuganyu96 (liboqs, 8 months, highest changes-requested rate, 176c inline median), rtjk (both repos, 10 months, consistent), Frauschi (both repos, highest per-comment inline median at 273c). Pairing each with baentsch or dstebila for a structured 3-month mentorship — explicit PR assignment, shadow-review, escalating responsibility — is the highest-ROI investment available.

P7

Investigate ashman-p review engagement

ashman-p has 73 events in the last 6 months across both repos and 35 months of tenure, but shows the lowest inline comment median (27c) and zero changes-requested across 64 approvals. Given the security-critical nature of the codebase, a direct conversation about review expectations and workload would be appropriate — this could reflect bandwidth constraints as much as disengagement, and the resolution matters either way.

P8

Publish a concrete post-NIST roadmap

The project's forward direction after NIST FIPS publication is not publicly articulated. A 12-month roadmap tied to open issues in GitHub Discussions would give prospective contributors a map of where help is needed and anchor the motivation of current contributors who joined for the standardization push. The January 2026 activity spike (37 contributors, 274 events) suggests continued external interest — a roadmap would convert that interest into directed contribution.

6. What-If Analysis — Key Maintainer Departure Scenarios

Given the concentration of activity in two individuals, this section quantifies the measurable impact on project activity if either or both were to leave. All figures are based on the last 6 months of data (Oct 2025–Mar 2026), bots excluded. The analysis measures only the direct loss of their contributions — it does not model secondary effects such as reduced mentorship capacity or slower PR review turnaround for others' work, which would amplify the impact further.

6.0a Activity Trend Warning: baentsch Post-LoA Decline

The what-if scenarios below may be closer than the headline numbers suggest. baentsch returned from the March 2025 leave of absence in September 2025 at 151 events that month. The post-LoA trend since then is a linear decline of approximately 20 events per month:

Month	baentsch events	3mo trailing avg	vs Sep 2025 peak
Sep 2025	151	151	baseline
Oct 2025	136	144	−10%
Nov 2025	68	118	−45%
Dec 2025	30	78	−80%
Jan 2026	40	46	−73%
Feb 2026	66	45	−56%
Mar 2026	20	42	−87%

At the observed linear trend of −20 events/month, extrapolated activity reaches zero within approximately 6 months of this report. This extrapolation is of course uncertain — activity is volatile and may recover. But the shape is structurally similar to the pre-departure trajectory of SWilson4, who maintained 40–75 events/month for 6 months (Jan–Jun 2025) before dropping to 1 event in July and zero thereafter. The key distinction is that baentsch has a documented LoA mechanism and has used it before, which makes a managed partial withdrawal more plausible than a sudden departure. Nevertheless, the decline is real and warrants direct attention.

Methodological caveat: This data cannot distinguish between burnout, deliberate role reduction, external time pressures, or temporary project lulls. The activity series measures GitHub engagement only. The claim is not that burnout has occurred, but that the observed trend is consistent with a pattern that has preceded departures in this dataset, and that the project's structural dependence on this one contributor makes the trend worth naming explicitly.

6.1 Current Baseline (Oct 2025–Mar 2026)

Month	Events (total)	Contributors	baentsch events	dstebila events
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2025-10	348	26	94	35
2025-11	241	29	51	20
2025-12	104	21	35	9
2026-01	274	37	47	33
2026-02	339	28	105	27
2026-03	173	17	28	6
6m total	1,479	avg 26.3/mo	360 (24%)	130 (9%)

6.0 A Note on Contributor Counts

The figure of 89 accounts active in the last 6 months requires qualification. Of those 89, 4 are bots and 53 are peripheral (single-event or single-month touches with no recurring engagement). The realistic community structure is:

Tier	Definition	Count	Who
Core	≥30 events across ≥3 months	8	baentsch, RodriM11, xuganyu96, dstebila, abhi-dev-engg, ashman-p, bhess, mkannwischer
Regular	≥10 events across ≥2 months	10	openforge-kt, rtjk, bbbbrumley, rben-dev, petrov, Frauschi, embetrix, HelenHZhang, M-AI-Noaimi, pi-314159
Occasional	≥5 events or review activity	14	ryjones, mhhkangg, woinan, bob-beck, hippy777, and 9 others
Peripheral	1–2 events, single month	53	One-time commenters, single PRs — not recurring participants

Effective active community: 18 people (Core + Regular). This aligns with direct observation of the repositories. The headline '89 active contributors' was technically accurate but deeply misleading — it counted anyone who left a single comment as active.

Current bus factor: 4. The top 4 contributors (baentsch 24%, RodriM11 13%, xuganyu96 10%, dstebila 9%) account for 57% of all activity. The 80% threshold is reached at contributor #11 (bbbbrumley). All 11 of those contributors are in the Core or Regular tier.

6.2 Scenario A: baentsch Leaves

Direct activity loss

- 360 events removed — 24% of 6-month total
- oqs-provider: 206 of 543 events (38%) — hardest hit
- liboqs: 154 of 936 events (16%)
- 33 of 102 approvals (32%) would need redistribution

Remaining capacity

- 1,119 events remain (76% of baseline)
- Avg contributors/month: 25.3 (minimal breadth impact)
- Top remaining reviewers: dstebila (38), abhi-dev-engg (36), xuganyu96 (33), RodriM11 (29)

- oqs-provider would lose its primary maintainer with no direct successor yet

Assessment: The overall ecosystem (liboqs) would absorb the loss with moderate disruption — contributor breadth is sufficient. oqs-provider faces a structural crisis: no currently active contributor has demonstrated the breadth of provider knowledge that baentsch holds. RodriM11 is the most plausible successor but has 12 months of tenure vs baentsch's 58.

6.3 Scenario B: dstebila Leaves

Direct activity loss

- 130 events removed — 9% of 6-month total
- liboqs: 111 of 936 events (12%)
- oqs-provider: 19 of 543 events (3%) — minimal direct impact
- 23 of 102 approvals (23%) would need redistribution

Remaining capacity

- 1,349 events remain (91% of baseline)
- baentsch (97), abhi-dev-engg (36), xuganyu96 (33) absorb review load
- liboqs community depth (37 contributors in Jan 2026) provides resilience
- Institutional knowledge loss is the primary risk, not activity volume

Assessment: dstebila's departure would be felt primarily as a loss of institutional depth — 114 months of project history — rather than immediate activity collapse. liboqs has sufficient contributor breadth to maintain operations. The co-founder's knowledge of historical design decisions, however, is not captured in any contributor's profile and would represent an irreplaceable loss for the project's long-term architectural direction.

6.4 Scenario C: Both Leave

Month	Baseline events	Without both	Retention %	Contributors
2025-10	348	185	53%	24
2025-11	241	158	66%	27
2025-12	104	74	71%	20
2026-01	274	184	67%	35
2026-02	339	252	74%	26
2026-03	173	136	79%	15
6m total	1,479	989 (67%)	—	avg 24.5/mo

Assessment: A 33% reduction in total activity with the remaining review load distributed across: abhi-dev-engg (36 review events), xuganyu96 (33), RodriM11 (29), bhess (25), ashman-p (24). The review layer exists but is thin — each of those five would need to roughly double their

review throughput to maintain current PR velocity. oqs-provider specifically would be at critical risk: no single active contributor has baentsch's provider breadth, and dstebila's liboqs oversight would also be absent.

6.5 What Would Mitigate These Risks

The what-if scenarios above are not predictions — they are planning tools. The mitigation actions that would change these outcomes most significantly:

- Formal co-maintainer designation for RodriM11 in oqs-provider before it is needed, with deliberate knowledge transfer (reduces Scenario A severity from critical to moderate)
- Documented architectural decision records (ADRs) capturing dstebila's design rationale in the repo itself — not in his head (reduces Scenario B institutional knowledge loss)
- Explicit PR assignment of xuganyu96 and abhi-dev-engg to liboqs review queues (builds the reviewer bench that Scenario C depends on)
- A written succession plan in GOVERNANCE.md naming the current maintainers, their areas, and their nominated successors — standard practice in mature open-source projects at this scale

7. Community Health Scorecard

The scorecard rates each dimension on a 1–10 scale based on current (last 6 months) data. Scores reflect community engagement health only, not code quality, security, or technical completeness.

Dimension	liboqs	oqs-provider	Weight	Notes
Contributor breadth	7.5	4.5	High	liboqs: 18 Core/Regular active; provider dominated by 1
Bus factor / resilience	6.0	3.0	High	liboqs BF=3 (excl. departed SWilson4); provider BF=2 nominal
Growth trend	7.0	4.0	Medium	liboqs stable post-NIST; provider down 54% from 2024 peak, now stabilising
New contributor intake	7.5	6.0	Medium	Both have new faces; retention still low
Retention rate	6.0	5.5	High	~22–25% at 3 months — structural challenge
Review culture	7.0	5.5	Medium	Inline substance improving; ashman-p concern noted
Cross-repo connectivity	6.0	4.5	Low	8 of 18 Core/Regular in both repos; 26 raw overlap inflated by peripheral accounts
Succession readiness	5.0	2.5	High	NEW: no documented succession; RodriM11 emerging
COMPOSITE SCORE	6.7	4.3		4-year window, excl. departed; succession dimension included

8. Governance Alignment — GOVERNANCE.md vs Activity Reality

A stale GOVERNANCE.md is worse than no GOVERNANCE.md. A project with no listed maintainer signals an open question. A project with a listed maintainer who has been inactive for 240 days signals a false reassurance — it tells users and potential contributors that stewardship is covered when the data shows otherwise. Both GOVERNANCE.md documents were read directly from the uploaded files and cross-referenced against the activity data. The framework uses three roles: Contributor, Committer (write access), and Maintainer (admin access), with a 90-day inactivity clause that automatically demotes Committers to Contributor status.

8.1 Current Named Role-Holders

Person	oqs-provider role	liboqs role	TSC	6-month activity
baentsch	Maintainer, Committer	Maintainer, Committer (leave of absence since Mar 11 2025)	Member	360 events — most active contributor in both repos despite LoA
dstebila	Release Manager	Maintainer, Committer	Member	130 events — Core tier
ashman-p	Maintainer, Committer	Committer	—	73 events — Core tier
bhess	Committer	Committer	Member	72 events — Core tier (4 in oqs-provider)
RodriM11	Committer	not listed	—	194 events — second most active; no liboqs role yet
xuganyu96	Release Manager	not listed	—	152 events in liboqs; no liboqs formal role
SWilson4	not listed	Maintainer	Past TSC	0 events since Jul 2025 — departed
praveksharma	not listed	Committer	—	0 events since Aug 2025 — departed
Martyrshot	not listed	Committer	—	0 events since Sep 2025 — >180 days
christianpaquin	Emeritus Committer	Committer	Member	2 events last 6m
vsoftco	Emeritus Committer	Committer	Member	1 event last 6m
abhi-dev-engg	not listed	not listed	—	109 events in liboqs; no formal role in either repo
zadlg	Emeritus	not listed	—	no recent activity (already Emeritus)

	Committer			
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Note: dstebila is Release Manager (not Committer) in oqs-provider — he can review but cannot merge PRs there. Merge authority in oqs-provider rests with baentsch, ashman-p, bhess, and RodriM11 only.

8.2 Discrepancies and Required Actions

oqs-provider is largely up to date. zadlg and christianpaquin are correctly listed as Emeritus; RodriM11 is correctly listed as Committer; xuganyu96 has the Release Manager role. The main gap is the structural single-Maintainer risk and missing succession documentation.

liboqs GOVERNANCE.md is significantly out of date. Three Committers and one Maintainer have been inactive for more than the 90-day threshold. The 90-day auto-downgrade rule has not been applied in any of these cases.

G1

liboqs: Remove SWilson4 from the active Maintainer list — a transparency obligation

SWilson4 is listed as Maintainer in liboqs. Last activity was July 2025 — over 240 days ago. Any user reading GOVERNANCE.md today sees a named Maintainer and reasonably concludes that someone is overseeing the project in that capacity. The data says otherwise. A blank maintainer list is preferable to a false one: it signals an open question rather than a false reassurance. The 90-day rule provides the mechanism; the obligation is to users who rely on GOVERNANCE.md to understand project health. Emeritus status is the appropriate designation if the departure is acknowledged by all parties.

G2

liboqs: Apply the 90-day rule to Martyrshot (Committer → Contributor)

Martyrshot is listed as Committer in liboqs. Last activity was September 2025 — over 180 days ago. The 90-day rule has unambiguously triggered. This is a mechanical update requiring no vote, only documentation.

G3

liboqs: Apply the 90-day rule to praveksharma (Committer → Contributor)

praveksharma is listed as Committer. Last activity was August 2025 — over 200 days ago. Same situation as Martyrshot: the 90-day rule has triggered automatically. The GOVERNANCE.md should be updated to reflect Contributor status.

G4

liboqs: Review borderline cases — christianpaquin and vsoftco

Both are listed as Committers in liboqs with minimal recent activity (2 and 1 events respectively in the last 6 months). Whether the 90-day rule has technically triggered depends on exact dates of their last substantive contribution. Both are active TSC members. The appropriate path is likely a voluntary Leave of Absence declaration (already available in the governance framework) to preserve voting rights while acknowledging reduced availability, rather than a forced downgrade.

G5

liboqs: Clarify baentsch's leave of absence status

baentsch declared a leave of absence as liboqs Maintainer/Committer on March 11, 2025,

yet is the single most active contributor across both repos with 360 events in the last 6 months. The leave of absence appears to apply specifically to formal decision-making and voting rights, while contribution activity continues. This ambiguity should be resolved in the GOVERNANCE.md with a clear statement of what the LoA covers and when it ends — the current wording implies suspended voting rights but the practice suggests full participation.

G6

liboqs: Initiate Committer vote for RodriM11

RodriM11 is already a Committer in oqs-provider with 194 events in the last 6 months spanning both repos. There is no formal role in the liboqs GOVERNANCE.md despite being the second-most active contributor. A Committer vote via GitHub Discussion — the process required by the document — would align formal standing with operational reality.

G7

liboqs: Consider Committer votes for xuganyu96 and abhi-dev-engg

xuganyu96 (152 events, highest changes-requested rate, Release Manager in oqs-provider) and abhi-dev-engg (109 events, 72 review events) are both operating at Committer-level engagement in liboqs without the formal role. xuganyu96 in particular has the most rigorous review profile in the dataset. Both contributors meeting the criteria stated in the document: 'sufficient code and deep subject matter knowledge in discussions'.

G8

oqs-provider: Initiate Maintainer vote for RodriM11

RodriM11 is already a Committer with 194 events in the last 6 months and demonstrated cross-repo engagement. The governance document allows any Committer to become a Maintainer by majority Committer vote. Initiating this vote now — before baentsch's workload becomes untenable — is the single most direct structural improvement available. A Maintainer designation gives RodriM11 the administrative access needed to step in during any baentsch absence.

G10

oqs-provider: Address ashman-p's Maintainer role explicitly

The data presents a genuine tension, but the transparency argument is unambiguous. A GOVERNANCE.md listing ashman-p as Maintainer tells users that this person is overseeing project health, leading communication, and defining technical guidelines. The activity data shows consistent presence (39 months, every month of the last 12, 27 PR merges) but a review profile that falls below what the GOVERNANCE.md explicitly requires of Committers — let alone Maintainers: 98% empty approval bodies, 27c median inline comments (lowest of any named role-holder in the dataset), and zero changes-requested across 64 approvals in 39 months. The GOVERNANCE.md states that Committers must 'review and merge pull requests.' Approving PRs with no substantive feedback is merge facilitation, not code review. The structural argument for retaining the Maintainer title is real — removing it leaves baentsch as sole Maintainer, which is worse. But this argument is about governance continuity, not accuracy. The resolution: elevate RodriM11 to Maintainer (G8), which makes the structural argument moot. At that point the question of ashman-p's appropriate role can be resolved honestly, with a direct conversation about review expectations that the data clearly warrants.

8.3 Governance Health Summary

Dimension	oqs-provider	liboqs	Status
Maintainer list accuracy	Current	Stale — SWilson4 inactive >240 days	G1 required
Committer list accuracy	Current	Stale — 3 inactive >90 days	G2/G3 required
90-day rule enforcement	Applied	Not applied — 3 Committers overdue	G1–G3 apply directly
Active contributors w/o role	None	3 (RodriM11, xuganyu96, abhi-dev-engg)	G6/G7 recommended
LoA documentation	N/A	Ambiguous (baentsch)	G5 recommended
Succession documentation	None needed*	Stale listing active	*honest gap > false reassurance
Release Manager role defined	Yes (2 named)	No (role not in liboqs doc)	Structural difference

9. Closing Assessment

liboqs has achieved something rare in open-source security infrastructure: genuine community breadth sustained over years. The 40-month streak of broad engagement, the record 2024 intake, and the resilient post-standardization activity levels all point to a project with a real community. The current active community of 18 people (8 Core, 10 Regular) includes a meaningful and formally recognised second tier — RodriM11 (Committer in oqs-provider), xuganyu96 (Release Manager in oqs-provider), abhi-dev-engg — none of whom had any formal role three years ago. The primary challenge is formalising the remainder of this second tier in liboqs, where RodriM11, xuganyu96, and abhi-dev-engg still lack formal roles despite operating at Committer-level engagement.

oqs-provider is at an inflection point that demands deliberate intervention. The combination of extreme maintainer concentration, a post-standardization decline, and a thin second-tier creates fragility that is invisible from the outside but unmistakable in the data. The good news: the raw material exists. RodriM11 is already a Committer, and bhess has the tenure to step into a larger role. The task is to formalise that path — a Maintainer vote for RodriM11 and the succession documentation recommended in G8 — before the need becomes urgent.

The ecosystem as a whole is well-positioned for the post-quantum era. The governance actions in section 8 — if implemented — address both the documentation debt (stale GOVERNANCE.md entries) and the recognition gap (active contributors without formal roles). Together with the what-if mitigations from section 6.5, they could realistically move the oqs-provider community health score from 4.3 toward 6.0+ within 12 months, not by manufacturing engagement, but by aligning formal structures with the activity that is already happening.

Report generated March 14, 2026 from GitHub interaction cache files (open-quantum-safe/liboqs and open-quantum-safe/oqs-provider). Data covers community engagement events only; commits, code review diffs, and CI results are not included. Analysis performed using the OQS GitHub Activity Analyzer.